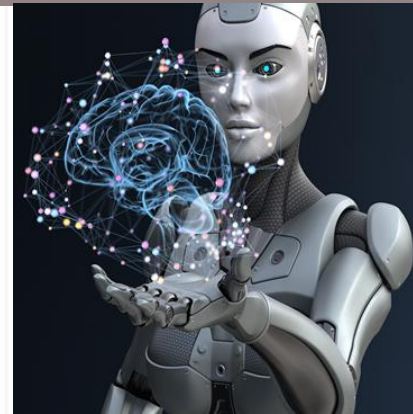
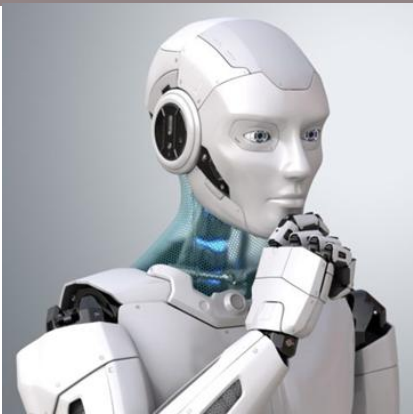


Artificial Intelligence



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Garry Kasparov vs IBM's Deep Blue



Out of 6 games, Kasparov won the 1st but IBM's Deep Blue defeated Kasparov by winning two games and drawing three (1996-97)



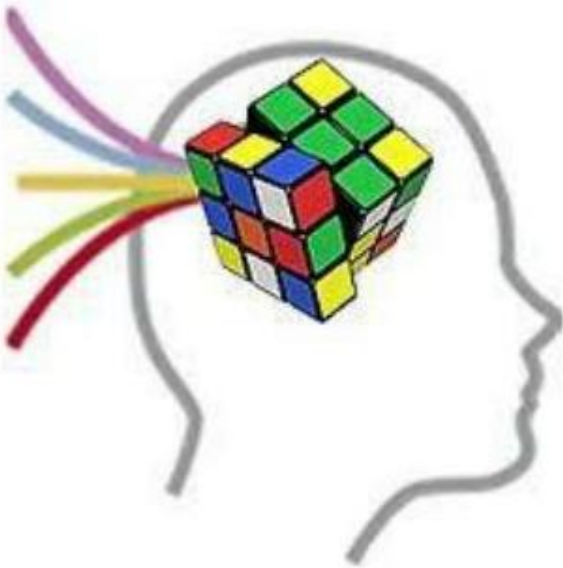
Sophia is a **social humanoid robot** developed by the Hong Kong-based company Hanson Robotics. (14th Feb 2016)

ChatGPT (Open AI)



- ChatGPT is a language model developed by OpenAI based on information available up to January 2022.
- It is designed to generate human-like text and engage in conversation with users.
- ChatGPT can understand and generate natural language text, making it suitable for a wide range of applications, including answering questions, providing explanations, generating text.
- ChatGPT is trained on a diverse range of internet text, which allows it to provide responses and carry on conversations in a coherent and contextually relevant manner.

What is Intelligence ?



- Ability to reason
- Ability to learn from experience
- Ability to apply knowledge to solve problems
- Ability to adapt to new situations
- ...

Intelligence: A Complex Cognitive Process

It encompasses a wide range of mental abilities and processes that involve the **acquisition**, **processing**, and **application of knowledge**.

1. Perception
2. Memory
3. Reasoning and Problem Solving
4. Language and Communication
5. Learning
6. Planning and Decision-Making
7. Abstract Thinking
8. Creativity
9. Emotional Intelligence
10. Social Intelligence

What is the Source of Intelligence ?

Intelligence can be attributed to many factors:

- Genetics
- Environmental factors
- Education
- Experience
- ..

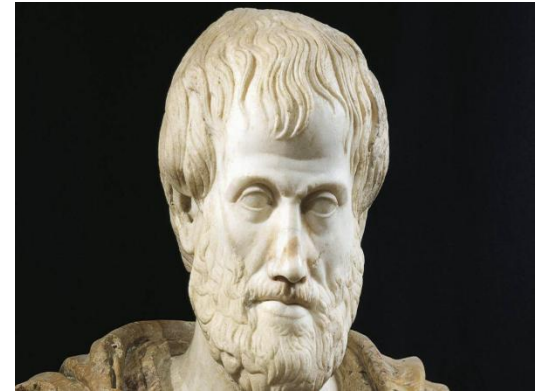


Origin of AI

Aristotle

[Greek Philosopher 384 BC to 322 BC]

“it would be nice for human to have
intelligent Machines to serve them”



Prof. John McCarthy

Dartmouth conference [1956]

“Every aspect of learning & human intelligence can be simulated in a machine”.



Formal Definitions

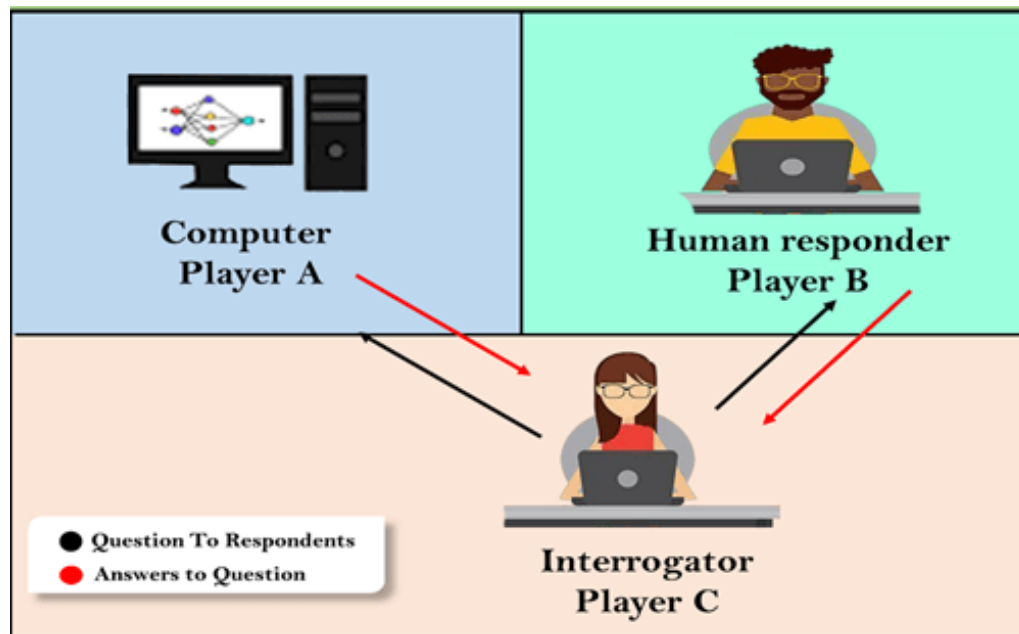
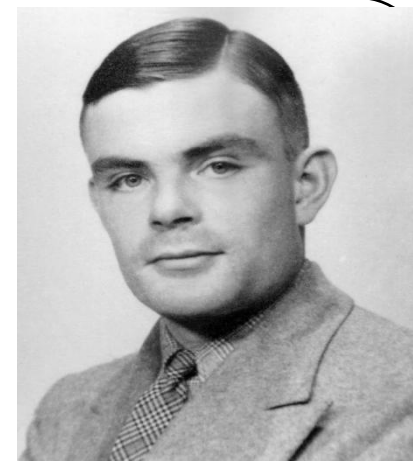
1. AI is the study of intelligent behavior.
[Geresreth Nilsson]
2. AI is the study of how to make computers to do things which at the moment people do better.
[Elain Rich]
3. AI is the field of science & engineering concerns with the understanding of what is commonly called intelligent behavior & with the creation of artifacts that exhibit such behavior.
[Shapiro]

Working definition & the goal

- **AI refers** to the simulation of human intelligence in machines. It involves creating algorithms/programs that enable computers to learn, reason, solve problems, and make decisions like humans.
- **The goal of AI** is to create intelligent machines that can assist, augment, or automate human tasks, making our lives more efficient and convenient.

Turing Test [Alan Turing]

British mathematician and computer scientist proposed the test in 1950.



Turing Test is a benchmark for evaluating AI's ability to mimic human thought process and behavior.

Turing Test (Contd..)

- **Human evaluator:** The test involves a human evaluator who interacts with both a human and a machine through a computer interface. The evaluator does not know which one is the machine and which is the human.
- **Conversation:** The evaluator engages in a text-based conversation with both the human and the machine. The goal is for the evaluator to determine which one is the machine.
- **Passing the Test:** If the evaluator is unable to distinguish between the machine & the human, the machine is said to have passed the Turing Test, implying that it exhibits human-like intelligence.

Turing Test (Contd..)

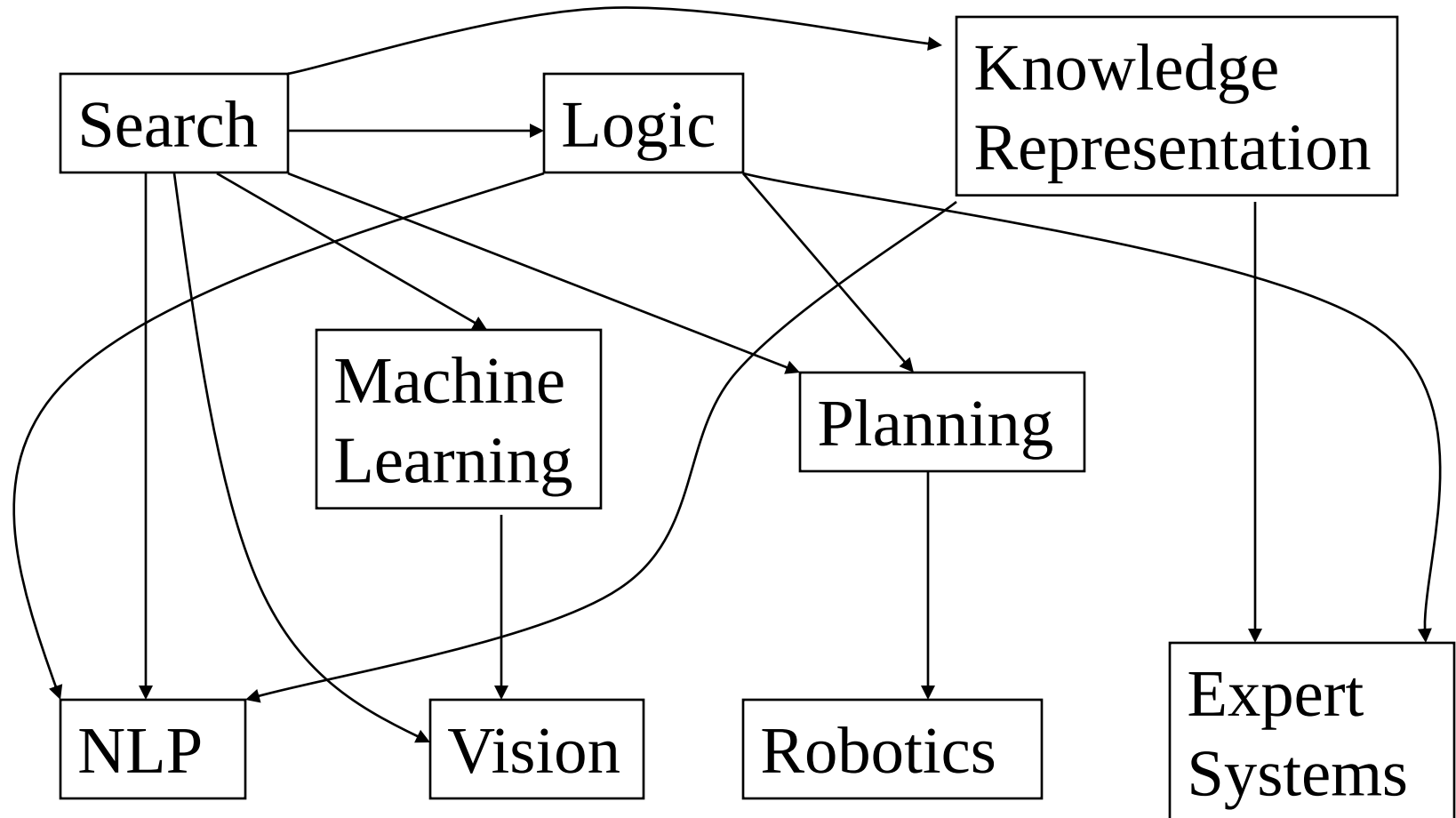
- If one can fool a human interrogator into believing that he/she is receiving answers from another human though it is actually from an AI system, then the AI system is said to have passed the Turing test of exhibiting human-like behavior.
- **ELIZA** was a Natural language processing computer program to demonstrate the ability of communication between computer and humans. It was one of the first **chatbots**, which was subjected to Turing Test.

Objective of AI:

To build intelligent systems that can exhibit human-like behavior in different situations.

- Sensing
- Learning rules, patterns & concepts
- Reasoning
- Deciding under uncertainty
- Planning strategies
- Understanding & interpreting events
- Searching

Major areas of AI



Applications of AI

- Disease diagnosis
- Fraud detection in finance
- Recommendation systems
- Predictive Maintenance of equipment
- Industrial robots
- Autonomous Vehicles
- Personalized Learning/ Virtual Tutors
- Content Creation
- Virtual Reality (VR) & Augmented Reality (AR)
- Chatbots and Virtual Assistants for customer service
- Biometric recognition
- Precision Farming
- Environment Monitoring